

Application of OOP Concepts in Codegnan University Management System

Below is a structured breakdown of **where each OOP concept is applied** in the system:

1. Encapsulation (Data Hiding)

- **Private Attributes:**
 - `self.__student_id` → Private Student ID
 - `self.__faculty_id` → Private Faculty ID
 - **Getter Methods:**
 - `get_student_id()` → Returns the **private student ID**
 - `get_faculty_id()` → Returns the **private faculty ID**
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2. Abstraction (Hiding Complex Implementation)

- **Abstract Method in Parent Class (**Person**)**
 - `display_info()` → Defined in **Person**, but not implemented
 - **Implemented in Child Classes:**
 - `Student.display_info()` → Displays **student-specific details**
 - `Faculty.display_info()` → Displays **faculty-specific details**
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3. Inheritance (Reusing Code from Parent Class)

- **Parent Class:**
 - **Person** (Attributes: `name`, `age`)
- **Child Classes:**
 - `Student(Person)` → Inherits `name`, `age` from **Person**
 - `Faculty(Person)` → Inherits `name`, `age` from **Person**
- **Using `super()` to Initialize Parent Attributes:**
 - `super().__init__(name, age)` → Used in **Student** and **Faculty** to call **Person's** constructor

4. Polymorphism (Same Method, Different Behavior)

- **Method Overriding:**
 - `Student.display_info()` → Displays **Student Details**
 - `Faculty.display_info()` → Displays **Faculty Details**
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5. Class & Instance Attributes

- **Class Attribute (Shared Across All Instances):**
 - `Person.university_name = "Codegnan University"`
 - **Instance Attributes (Unique to Each Object):**
 - `self.name, self.age, self.course` (for `Student`)
 - `self.department` (for `Faculty`)
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6. Class Methods, Instance Methods, and Static Methods

- **Instance Method (Works with Object Attributes):**
 - `display_info()` (Overridden in `Student` and `Faculty`)
- **Class Method (Works with Class-Level Attributes):**
 - `@classmethod total_students(cls)` → Returns total students
 - `@classmethod total_faculty(cls)` → Returns total faculty
- **Static Method (Utility Function, Doesn't Require Class or Instance Data):**
 - `@staticmethod university_motto()` → Returns university motto